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POST: Prototype-oriented similarity transfer framework for cross-domain facial expression recognition

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Abstract

Facial expression recognition (FER) is one of the popular research topics in computer vision. Most deep learning expression recognition methods perform well on a single dataset, but may struggle in cross-domain FER applications when applied to different datasets. FER under cross-dataset also suffers from difficulties such as feature distribution deviation and discriminator degradation. To address these issues, we propose a prototype-oriented similarity transfer framework (POST) for cross-domain FER. The bidirectional cross-attention Swin Transformer (BCS Transformer) module is designed to aggregate local facial feature similarities across different domains, enabling the extraction of relevant cross-domain features. The dual learnable category prototypes is designed to represent potential space samples for both source and target domains, ensuring enhanced domain alignment by leveraging both cross-domain and specific domain features. We further introduce the self-training resampling (STR) strategy to enhance similarity transfer. The experimental results with the RAF-DB dataset as the source domain and the CK+, FER2013, JAFFE and SFEW 2.0 datasets as the target domains, show that our approach achieves much higher performance than the state-of-the-art cross-domain FER methods.

KEYWORDS

bidirectional cross-attention, facial expression recognition, learnable category prototypes, similarity transfer

1 | INTRODUCTION

Automatic facial expression recognition (FER) is one of the hot research topics in the field of computer vision, which is widely used in human-computer interaction, security, robotics manufacturing, medical diagnosis, communication and other fields.[1] The existing FER methods can be divided into two categories, one is the traditional FER method, and the other is the deep learning method. Traditional methods manually extract features after data pre-processing, and then use machine or non-machine learning models for classification; while deep learning methods integrate two stages of feature extraction and classification to automatically realize facial expression recognition.[2] Since deep learning models have extremely strong feature extraction ability, which enables computers to deeply understand the facial expression, they are widely accepted by scholars and have become the mainstream method for FER. After more than ten years of development, FER based on deep learning has made revolutionary progress in network structure design and model optimization.[2] While early FER studies focused on the performance of single datasets, particularly lab-controlled datasets, challenges emerged when researchers attempted to generalize their work across wider range of application scenarios.[3,4] The differences in

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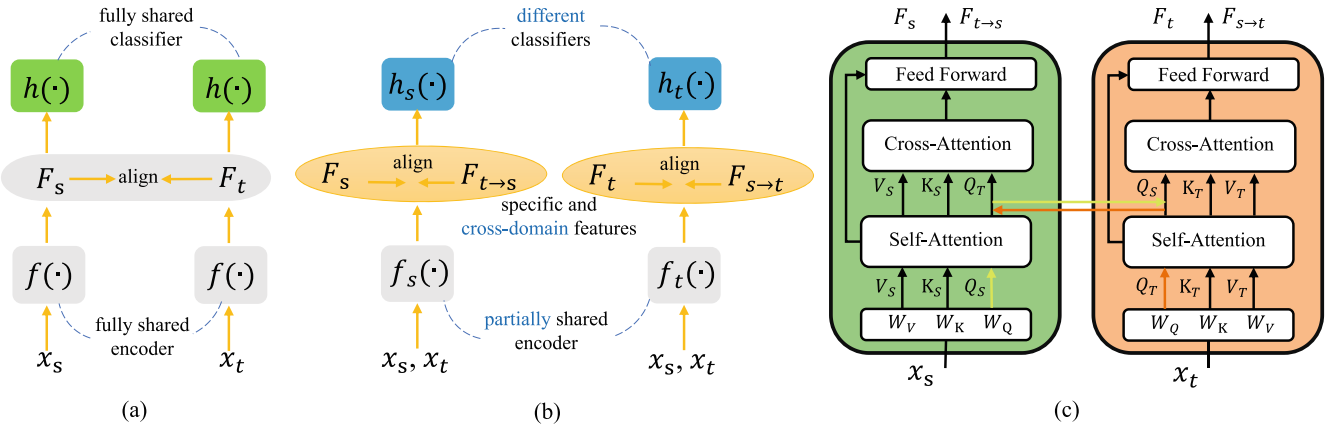


FIGURE 1 Bidirectional cross-attention swin transformer module. (a) The current CD-FER paradigm, (b) our designed CD-FER paradigm, (c) the structure of our proposed bidirectional cross-attention swin transformer module.

data capture environments, along with variations in the subjects' ethnic backgrounds, introduced significant dataset heterogeneity. Furthermore, the subjective nature of facial expression annotations exacerbated inter-domain differences.[2] Consequently, methods that performed well within a single dataset may struggled when applied to diverse datasets.[5]

In recent years, researchers have become increasingly interested in the field of cross-domain facial expression recognition (CD-FER). CD-FER strives to build domain-invariant representations to enhance cross-domain generalization, paying special attention to scenarios where the target domain lacks labeled data. A dominant trend among these algorithms is to utilize convolutional neural networks (CNNs) to extract domain-specific and domain-invariant features, and then employ maximum mean discrepancy (MMD)[6] to reduce feature distribution differences, achieving cross-domain alignment.

However, a common limitation of these methods is their overemphasis on domain-specific information during feature extraction, which can result in overlooking local content similarities across datasets. Despite the differences in data distributions between datasets, it is worth noting that local facial features often exhibit noteworthy similarities. As a result, these local similarities contain valuable information crucial for achieving fine-grained adaptation.

To leverage these local similarities and mitigate feature distribution deviation, we introduce the bidirectional cross-attention module within the swin transformer (ST),[7] which we refer to as the bidirectional cross-attention Swin Transformer (BCS Transformer). As illustrated in Figure 1c, this module is specifically designed to aggregate local facial feature similarities across different domains, enabling the extraction of relevant cross-domain features–something that traditional CNNs struggle to achieve.

Furthermore, to address the issues of degradation of target domain discriminator and unsatisfactory decision boundary in current CD-FER paradigm, illustrated in Figure 1a, we introduce dual learnable category prototypes to represent potential space samples for both source and target domains. These prototypes serve as domain-specific classifiers, mitigating the drawbacks of a universally shared classifier. Continuously refined through back-propagation, these prototypes more effectively align with the cross-domain and domain-specific features extracted by the BCS Transformer, thereby improving similarity transfer, as illustrated in Figure 1b.

In order to maximize the utilization of local facial feature similarities across domains and alleviate category imbalance, we introduce an end-to-end training strategy: the self-training resampling (STR) algorithm, enhancing the recognition capabilities of the model.

Finally, we propose a prototype-oriented similarity transfer framework for CD-FER, named POST, as shown in Figure 2. Our POST takes the source and target domains as inputs, the framework leverages the BCS Transformer to derive both cross-domain and domain-specific features. Subsequently, these features inform the calculation of the loss function, facilitating domain alignment and the updating of domain prototypes. The processed enhanced images are reintegrated into the source domain using the STR strategy, which is used to enhance the relevance of the source to the target domain.

Overall, our main contributions can be summarized as follows:

1. We integrate a bidirectional cross-attention module within the swin transformer to consolidate the similarity of local facial features across domains, yielding distinct cross-domain features.

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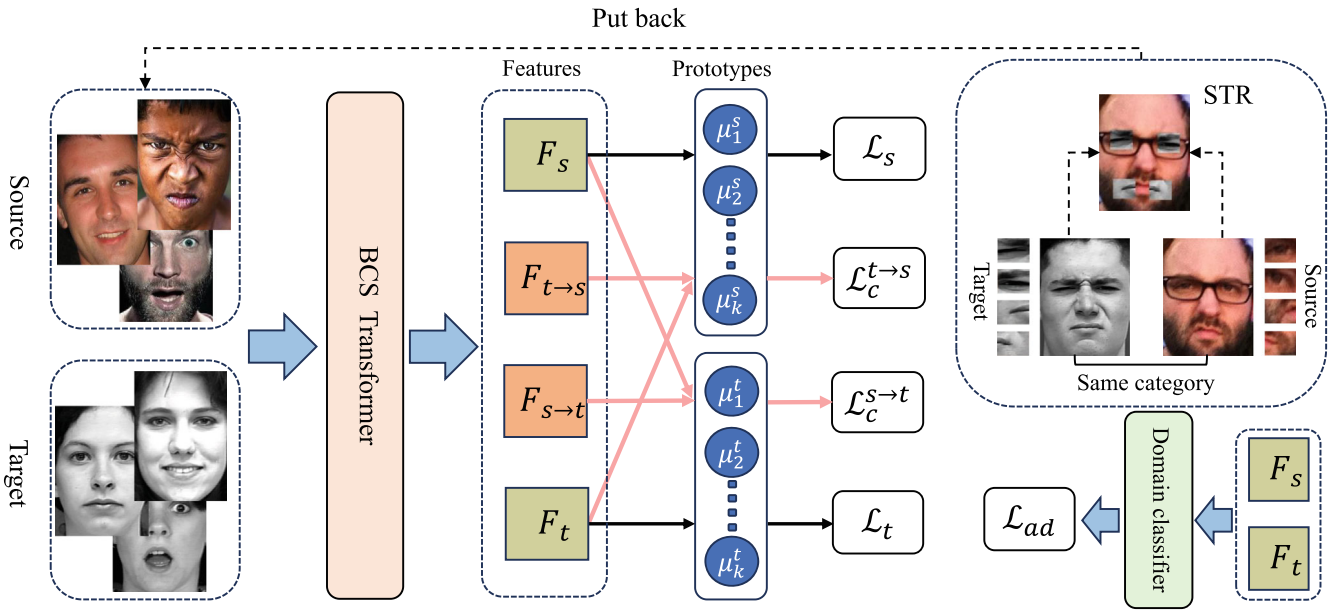


FIGURE 2 The framework of our proposed prototype-oriented similarity transfer (POST) CD-FER method.

2. We propose a novel CD-FER framework that incorporates dual learnable category prototypes, ensuring enhanced domain alignment by leveraging both cross-domain and specific domain features. Additionally, we introduce an end-to-end STR strategy to improve similarity transfer for CD-FER.
3. We evaluate the proposed method on CK+,[8] FER2013,[9] JAFFE[10,11] and SFEW2.0[12] datasets, using the RAF-DB[13] dataset as the source domain, and achieve state-of-the-art performance in terms of recognition rate.

2 | RELATED WORK

2.1 | Transformer for vision

The Transformer architecture has achieved remarkable success.[14] Inspired by these advancements, researchers have explored Transformer architectures as potential alternatives to traditional CNN backbones in computer vision. The vision transformer (ViT)[15] notably uses image patches as input into a transformer encoder to tackle image classification tasks. Recently, the swin transformer (ST),[7] pyramid vision transformer (PVT),[16] and CrossViT[17] further refine the architecture from various angles. The ST, in particular, employs mobile window-based self-attention to minimize computational complexity, coupled with a hierarchical design inspired by CNNs to effectively handle images of varying sizes.

Since Transformer has excellent capability of transferable representation, several studies et al.[18,19] have utilized it to address the challenge of domain adaptation. Cross-domain transformer (CDTrans)[19] incorporates the entire ViT architecture and introduces a cross-attention mechanism from target to source domain for selected pairs. Different from CDTrans, our approach recognizes the potential of bidirectional cross-attention between the source and target domains to comprehensively merge divergent feature spaces. By embedding bidirectional cross-attention into the ST model, we enable the extraction of cross-domain features that combine insights from both domains. These cross-domain and domain-specific features then inform the prototypes of both domains, enhancing domain alignment for improved performance.

2.2 | Cross-domain facial expression recognition

Traditional FER algorithms commonly experience performance degradation in cross-dataset evaluations due to the inherent subjectivity in the annotation process and variations in data collection conditions.[3,4] These discrepancies can lead to challenges in generalizing the learned patterns to new datasets, resulting in noticeable drops in performance.

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In recent years, a large number of CD-FER algorithms have emerged to address this challenge. The most prominent are those that utilize MMD[6] to achieve invariant subspace. Besides, inspired by generative adversarial networks (GAN),[20] techniques[21,22] that deploy adversarial training, aiming to distill domain-independent representations through a strategic min-max game with a domain discriminator.

A deeper insight into the disparities between facial expression datasets, as elucidated in Reference 23, reveals differing conditional probabilities across datasets. To address this problem, emotion-conditional adaption network (ECAN)[23] has been proposed, which not only learns domain-independent features but also seeks to align both the marginal and conditional probability distributions across domains, all while accommodating domain class imbalances. In the field of graph representation learning, Chen et al.[24] innovatively incorporated adversarial training to craft the adversarial graph representation adaptation (AGRA) framework. Recently, Wang et al.[25] integrated contrast learning into CD-FER to optimize the similarity metrics for positive and negative sample pairs.

Despite the significant progress in CD-FER, a common limitation persists: many existing methods excel at extracting domain-specific features but often overlook the subtle local content similarities present across datasets. This oversight can hinder the ability to capture nuanced facial expressions that may be shared between different domains, impacting the overall performance of the recognition system.

3 | METHOD DESCRIPTION

For the CD-FER task, there is a labeled source dataset $D_s = \{x_s^i, y_s^i\}_{i=1}^{N_s}$, with marginal probability distribution $P_s(X)$ and an unlabeled target dataset $D_t = \{x_t^i\}_{i=1}^{N_t}$ with marginal probability distribution $P_t(X)$, where N_s and N_t are the number of images in source and target domains respectively. The two datasets have different distributions ($P_s(X)! = P_t(X)$). Our main work focuses on developing a robust CD-FER framework using labeled data in the source dataset and unlabeled queries from the target dataset. As a solution, we propose a prototype-oriented similarity transfer framework named POST, as shown in Figure 2. Our POST leverages the BCS Transformer to derive both cross-domain and domain-specific features. Subsequently, these features inform the calculation of the loss function, facilitating domain alignment and the updating of domain prototypes. Furthermore, an end-to-end universal STR strategy is introduced to reintegrate enhanced images into the source domain. Our POST is dedicated to bridge and reduce the domain disparities between source and target domains and ultimately improving facial expression recognition accuracy in the target domain.

3.1 | BCS transformer

In the field of FER, the subtle differences in local kernel regions of the face, such as eyes and corners of the mouth, are critical to the recognition results. Despite the observable distributional variances across different datasets, there is a noticeable similarity in the dynamics of these local features. To better leverage this inherent similarity, we propose the BCS Transformer. This approach integrates a bidirectional cross-attention module in the ST, which is already recognized for its exceptional performance in various computer vision tasks. A key component of the ST is the window-based self-attention module, which further enhances the model's ability to capture and integrate relevant local features for improved facial expression recognition accuracy. In ST, the image $I \in \mathbb{R}^{H\times W\times C}$ is reshaped into a sequence of flattened 2D patches $r \in \mathbb{R}^{N\times(P^2\times C)}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch and different in every stage, and $N = HW/P^2$ is the resulting number of patches. For self-attention, the patches are first projected into three vectors, that is, queries $Q \in \mathbb{R}^{N\times d_k}$, keys $K \in \mathbb{R}^{N\times d_k}$ and values $V \in \mathbb{R}^{N\times d_v}$. d_k and d_v are their dimensions. The queries are matched with the keys to get a $N \times N$ self-attention matrix whose elements represent the semantic relevance of the corresponding query-key pairs. Conclusively, utilizing the self-attention matrix, new embedding values can be derived as weighted sums of the respective values. For each distinct domain, the computation of their self-attention module can be described as follows:

$$Attn_s(Q_s, K_s, V_s) = softmax\left(\frac{Q_s K_s^T}{\sqrt{d_k}}\right)V_s \tag{1}$$

$$Attn_t(Q_t, K_t, V_t) = softmax\left(\frac{Q_t K_t^T}{\sqrt{d_k}}\right)V_t \tag{2}$$

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where Q_s, K_s, and V_s represent three vectors for source domain, and Q_t, K_t, and V_t represent three vectors for target domain.

The cross-attention module serves as an extension of the self-attention mechanism, with a main distinction being the origin of keys and values. In self-attention, keys and values are derived from the same domain, while in cross-attention, they come from different domains. When applied to an image containing multiple patches, the patch that is most similar to the query from another image, sourced from a separate domain, is given a higher weight, leading to a more substantial impact on the output. Essentially, the output of the cross-attention module combines information from two input images, highlighting their shared or similar patches and facilitating the integration of cross-domain features for improved facial expression recognition performance. The computation for the cross-attention module can be formulated as follows:

$$Attn_{t\to s}(Q_t, K_s, V_s) = softmax\left(\frac{Q_t K_s^T}{\sqrt{d_k}}\right)V_s \tag{3}$$

$$Attn_{s\to t}(Q_s, K_t, V_t) = softmax\left(\frac{Q_s K_t^T}{\sqrt{d_k}}\right)V_t \tag{4}$$

Images from the source and target domains are processed using the self-attention module, resulting in domain-specific feature representations denoted as F_s for the source and F_t for the target. Using the cross-attention module, we can further derive cross-domain features, represented as $F_{t\to s}$(from target to source) and $F_{s\to t}$(from source to target). Here, F symbolizes an m-dimensional feature vector.

3.2 | Learnable category prototypes

In unsupervised classification tasks, the k-means clustering algorithm is commonly employed to determine category prototypes. However, this technique requires significant computational resources, and a crucial limitation is its inability to promptly update category prototypes in response to changes in model parameters. Such limitations can hinder the efficacy of CD-FER. Inspired by Tanwisuth et al.,[26] we propose a method to parameterize the prototypes for both source and target domains. Our method allows the prototypes to be dynamically learned through the backpropagation algorithm, potentially enhancing domain feature alignment and ensuring a more adaptive response to changes.

Specifically, we define the prototype of every class as a vector $\mu \in \mathbb{R}^m$, where m is the dimension of each prototype that is the same as the embedding dimension of the sample output feature F. All prototypes form two domain's prototype matrices $M_s = \{\mu_s^1, \mu_s^2, \ldots, \mu_s^k\}$ and $M_t = \{\mu_t^1, \mu_t^2, \ldots, \mu_t^k\}$, where k is the number of categories. We can compute similarity matrix vector between F and M_s as $h_s(F^i) = [p_s^{i,1}, p_s^{i,2}, \ldots, p_s^{i,k}]$, with $p_s^{i,j} = \frac{\exp(\mu_s^j \cdot F^i)}{\sum_{r=1}^k \exp(\mu_s^r \cdot F^i)}$. Similarly, we can get the similarity matrix of features to the target domain prototype $h_t(F^i) = [p_t^{i,1}, p_t^{i,2}, \ldots, p_t^{i,k}]$, with $p_t^{i,j} = \frac{\exp(\mu_t^j \cdot F^i)}{\sum_{r=1}^k \exp(\mu_t^r \cdot F^i)}$. We learn source and target prototypes by minimize cross entropy loss function:

$$\mathcal{L}_s = \frac{1}{N_s}\sum_{i=1}^{N_s}\mathcal{L}_{CE}(h_s(F_s^i), y_s^i) \tag{5}$$

$$\mathcal{L}_t = \frac{1}{N_t^*}\sum_{i=1}^{N_t^*}\mathcal{L}_{CE}(h_t(F_t^i), \hat{y}_t^i) \tag{6}$$

And the loss for domain-specific prototypes learning is:

$$\mathcal{L}_{in} = \mathcal{L}_s + \mathcal{L}_t \tag{7}$$

Since the target domain has no labels, we generate pseudo-labels as $\hat{y}_t$ (we use hard labels) for the target domain by using k-means clustering algorithm every several epochs. We only use pseudo-labels for samples whose predictive labels are the same as pseudo-labels and predictive confidence is higher than a certain threshold, denoted as $D_t^{trust} = \{x_i\}_{i=1}^{N_t^*}(x_i \in D_t, cnf(x_i) > thr, pre(x_i) = pse(x_i))$. The prototypes of the target domain are initialized to the pre-trained prototypes of the source domain.

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3.3 | Domain alignment

Compared with single-domain features, cross-domain features contain not only the information of the source domain but also the information of the target domain. As a result, we use cross-domain features to perform supervised learning to achieve the purpose of domain feature alignment. In addition, we calculate matrix of cross-similarity of domain features and domain prototypes. Specifically, we compute similarity matrix between $\{F_{t\to s}, F_t\}$ and source domain prototype M_s as $\left\{h_s(F^i_{t\to s}), h_s(F^i_t)\right\}$. In the same way, we can also calculate $\{F_{s\to t}, F_s\}$ and target domain prototype M_t as $\left\{h_t(F^i_{s\to t}), h_t(F^i_s)\right\}$. Then we use the cross-entropy loss function to maximize the similarity between features and prototype:

$$\mathcal{L}_{t\to s} = \frac{1}{N_s}\sum_{i=1}^{N_s}\mathcal{L}_{CE}(h_s(F^i_{t\to s}), y^i_s) + \frac{1}{N^*_t}\sum_{i=1}^{N^*_t}\mathcal{L}_{CE}(h_s(F^i_t), \hat{y}^i_t) \tag{8}$$

$$\mathcal{L}_{s\to t} = \frac{1}{N^*_t}\sum_{i=1}^{N^*_t}\mathcal{L}_{CE}(h_t(F^i_{s\to t}), \hat{y}^i_t) + \frac{1}{N_s}\sum_{i=1}^{N_s}\mathcal{L}_{CE}(h_t(F^i_s), y^i_s) \tag{9}$$

Additionally, we employ an adversarial domain adaptation approach that incorporates a discriminator to determine the source domain of a given sample. This setup serves two purposes: firstly, it encourages the feature extractor to deceive the domain discriminator maximally, making it challenging to distinguish which domain the samples come from. Secondly, the domain discriminator is trained to optimally distinguish between features of the source and target domains. Through this adversarial competition, the feature extractor is fine-tuned to extract features that are substantially similar across both source and target domain data, thus promoting domain invariance.

$$\mathcal{L}_{ad} = \frac{1}{N_s}\sum_{i=1}^{N_s}\log\frac{1}{D(F^i_s)} + \frac{1}{N_t}\sum_{i=1}^{N_t}\log\frac{1}{1-D(F^i_t)} \tag{10}$$

The final loss function for domain alignment is:

$$\mathcal{L}_{cross} = \alpha\mathcal{L}_{t\to s} + \beta\mathcal{L}_{s\to t} + \gamma\mathcal{L}_{ad} \tag{11}$$

3.4 | Self-training resample

Imbalances in data distribution are common in facial expression datasets and manifest in most real-world scenarios. Such disparities can be attributed to the practical complexity of data collection. For instance, collecting and annotating instances of happy is relatively straightforward; in contrast, gathering data for expressions like disgust, angry, or other less frequent facial expressions proves to be much more difficult. The domination of one category, coupled with the scarcity of another, may affect the relevance of the source domain with respect to the target task. To address this issue, we introduce a self-training resampling strategy. Specifically, we identify the category with the least representation based on the model's prediction labels for the target domain, designating it as the augmented category of our interest. Then select the target domain image whose prediction threshold is greater than the resampling threshold thr_r and that falls into the category y^*, denoted as $D^r_t = \{x_i\}^{N_t}_{i=1}(x_i \in D_t, cnf(x_i) > thr, pre(x_i) = y^*)$. We then select images of the same category in the source domain as $D^r_s = \{x_i\}^{N_s}_{i=1}(x_i \in D_s, y_i = y^*)$. Finally, we replace the four facial features slices (left eye, right eye, left mouth corner, and right mouth corner) of the image in D^r_s with D^r_t pair one by one. The processed image should be placed into the target domain dataset. Resampling is performed once for several epochs. Algorithm 1 summarizes the processes of our proposed STR. Experiments have shown that this strategy has achieved excellent results.

4 | EXPERIMENTS

4.1 | Datasets

The CK+[8] dataset is collected using the same video equipment under laboratory conditions, and then frame extracted image data. Each subject performed 23 facial displays, including a total of eight basic expressions.

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Algorithm 1. Self-training resample (STR)

Input: Source data with labels $D_s = \{x_s^i, y_s^i\}_{i=1}^{N_s}$ and target data with no labels $D_t = \{x_t^i\}_{i=1}^{N_t}$.
Output: Augmented source dataset.

```
for i to N_s do
    Calculate the quantity of each category count(y_s^t)++ in source data;
end
Select the category y* with the fewest number;
for i to N_t do
    Using model calculates the predictive logical values pre(x_t^i) and label pse(x_t^i) of the target domain image;
    if pre(x_t^i) > thr and pse(x_t^i) = y* then
        random select x'_s ∈ D_s with y'_s = y*;
        for landmark = Left eye, right eye, corner of left mouth, corner of right mouth do
            x'_s[landmark] = x_t^i[landmark];
            put back x'_s with label y* to source dataset;
            update number of category count[y*].
        end
    end
end
```

The FER2013[9] dataset is a competition dataset released by Google on the Kaggle platform in 2013. The images are manually labeled by trained human coders, including seven expressions and a total of 35,887 images.

The JAFFE[10,11] dataset collected seven basic facial expressions of 10 Japanese female college students under laboratory conditions, with a total of 213 images, all of which were grayscale images.

The SFEW2.0[12] dataset is a static image dataset extracted from movies, with a total of 958 training images and 436 validation images.

The RAF-DB[13] dataset is a real-world dataset collected from the internet. The dataset contains 15,339 images from thousands of individuals.

4.2 | Implementation details

For all images, the MTCNN[27] is first used to locate the facial regression frame and the five facial key points, and the images in which faces cannot be detected are removed from the dataset. Subsequently, faces are intercepted from the background using the facial regression frame. Images with the resolution below 20×20 are removed, and the remaining images containing only faces are cropped 224×224 and used as inputs to the model. Furthermore, to improve the robustness of the model, images are also augmented by flipping horizontally with 0.5 probability. We set the base learning rate as 0.001 for each target domain. We train for 256 epochs using stochastic gradient descent (SGD) with momentum 0.9 and weight decay 0.001. We use cosine descent to adjust the learning rate. The batch-size is 64 and we set the resampling threshold to 0.95 and the number of resamples to 10 for both domains. All experiments are implemented by PyTorch, and run on a PC with a NVIDIA GTX 4090 GPU.

4.3 | Comparison with the state-of-the-art cross-domain FER methods

In this section, we compared our proposed POST method with other methods on four FER target datasets using RAF-DB as the source dataset. As shown in Table 1, it can be seen that our proposed POST performes well on all target datasets, with significantly higher mean recognition rates than other comparison methods. The CK+ dataset, which was captured under laboratory conditions, has good data quality and many methods have achieved promising results. Our POST achieves the highest accuracy of 92.51%, which is 6.02% higher than the next best result. The JAFFE dataset, although also in laboratory conditions, is inherently biased in terms of gender because it only covers ten Japanese women. Our POST is

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TABLE 1 Comparison of FER rates (%) on CK+, JAFFE, SFEW2.0, and FER2013 datasets.

Method	Source dataset	Target dataset				
		CK+	JAFFE	SFEW2.0	FER2013	Mean
CADA[22]	RAF-DB	72.09	52.11	53.44	57.61	58.81
LPL[13]	RAF-DB	74.42	53.05	48.85	55.89	58.06
SAFN[28]	RAF-DB	75.97	61.03	52.98	55.64	61.41
DETN[21]	RAF-DB	78.22	55.89	49.40	52.29	59.85
JUMBOT[29]	RAF-DB	79.46	54.13	51.97	53.56	59.78
AGRA[24]	RAF-DB	85.27	61.50	56.43	58.95	65.54
ECAN[23]	RAF-DB	86.49	61.94	54.34	58.21	65.25
POCAN[25]	RAF-DB	84.50	64.32	53.44	58.87	65.28
POST(Ours)	RAF-DB	**92.51**	**69.05**	**56.69**	**64.92**	**70.79**

Note: The bold indicates optimal.



FIGURE 3 Samples in the SFEW 2.0 dataset that are easy to misidentify. The text on the image represents the ground truth of the category.

also able to achieve a 69.05% accuracy rate. Regarding the more eclectically sourced datasets, FER2013 and SFEW 2.0, our POST consistently outperforms other methods. Specifically, on the SFEW 2.0 dataset, our advantage is only 0.26% of that of Reference 24, which may be due to the obvious domain differences and the limited similarity data at the transfer stage. The FER2013 dataset contains a high number of mislabels, which is a greater challenge to get a high FER rate. However, our POST still obtains an accuracy of 64.92%, which is 5.97% higher than the next best method.[24] Overall, our POST achieves an average accuracy of 70.79% across the four target datasets, which far exceeds the other compared methods.

In addition, we also compared our POST on the SFEW 2.0 dataset, which contains more confusing (compound) expression data, with the ARGA method,[24] which also performs well on this dataset. While Table 1 shows that the recognition rates of our POST on SFEW 2.0 are very close to those of ARGA, the performance of our method on complex expression recognition has a clear advantage as shown by the recognition results for the hard samples of ambiguous expression data given in Figure 3 and Table 2. As can be seen from Table 2, for hard samples of complex expressions with ambiguity, our POST can accurately determine the categories, while ARGA makes classification errors. This demonstrates the robustness of our method in dealing with subtle compound expressions.

Furthermore, we also gave the variation of the recognition rate of our POST on each target dataset with the training process, as shown in Figure 4. From the results, it can be seen that for the CD-FER task, despite the obvious

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TABLE 2 Comparison of recognition results for the hard samples of ambiguous expression.

Hard samples	Ground truth	Method	
		ARGA[24]	Ours
Figure 3a	Angry	Disgust	Angry
Figure 3b	Sad	Happy	Sad
Figure 3c	Happy	Neutral	Happy
Figure 3d	Surprise	Fear	Surprise
Figure 3e	Neutral	Sad	Neutral
Figure 3f	Angry	Happy	Angry

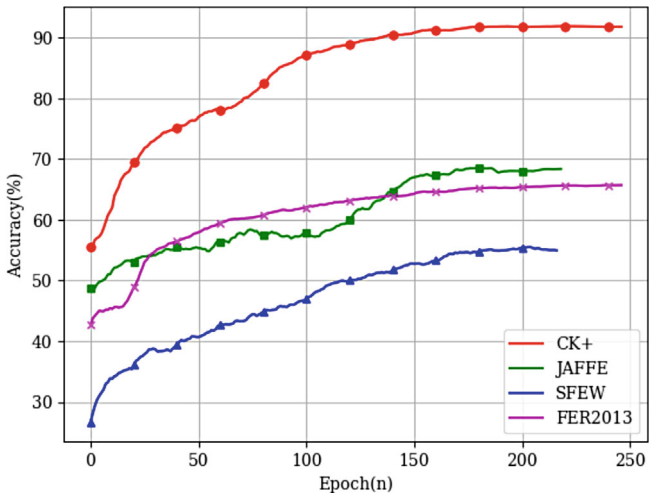


FIGURE 4 The FER rates during the training process on four target datasets.

deviations in the distribution of facial expression features in different target datasets, there are still commonalities in facial features. Utilizing these common attributes can enhance the consistency of domain features and thus improve the accuracy of CD-FER.

4.4 | Ablation study

In this section, we conducted comprehensive experiments to analyze the effectiveness of the various components of our proposed POST framework.

4.4.1 | Verify the effects of components and strategies

The POST framework consists of three crucial components: the domain-specific prototype learning $\mathcal{L}_{in}$, the domain feature alignment $\mathcal{L}_{cross}$, and the self-training resampling module (STR). To validate the contribution of each component to the POST model, we conducted an ablation study across four target datasets. As a baseline, we utilized the BSC Transformer, equipped only with the losses ($+\mathcal{L}_{in}$) designed to discriminate between source and target domain class prototypes. The experimental results are detailed in Table 3.

After integrating the domain alignment loss ($+\mathcal{L}_{in}+\mathcal{L}_{cross}$) for cross-domain knowledge acquisition, we notice a significant improvement in the mean accuracy of FER, which jumped from 65.86% to 67.62%. This highlights the value of the proposed cross-domain feature loss in relation to the specific domain class prototype. Such an alignment assists the model

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TABLE 3 Evaluation of the effectiveness of each modules on different datasets with FER rates (%).

Method	$\mathcal{L}_{in}$	$\mathcal{L}_{cross}$	STR	CK+	JAFFE	SFEW2.0	FER2013	Mean
POST	✓	—	—	87.50	65.04	52.71	58.18	65.86
	✓	✓	-	89.24	67.39	53.44	60.37	67.62
	✓	✓	✓	**92.51**	**69.05**	**56.69**	**64.92**	**70.79**

Note: The bold indicates optimal.

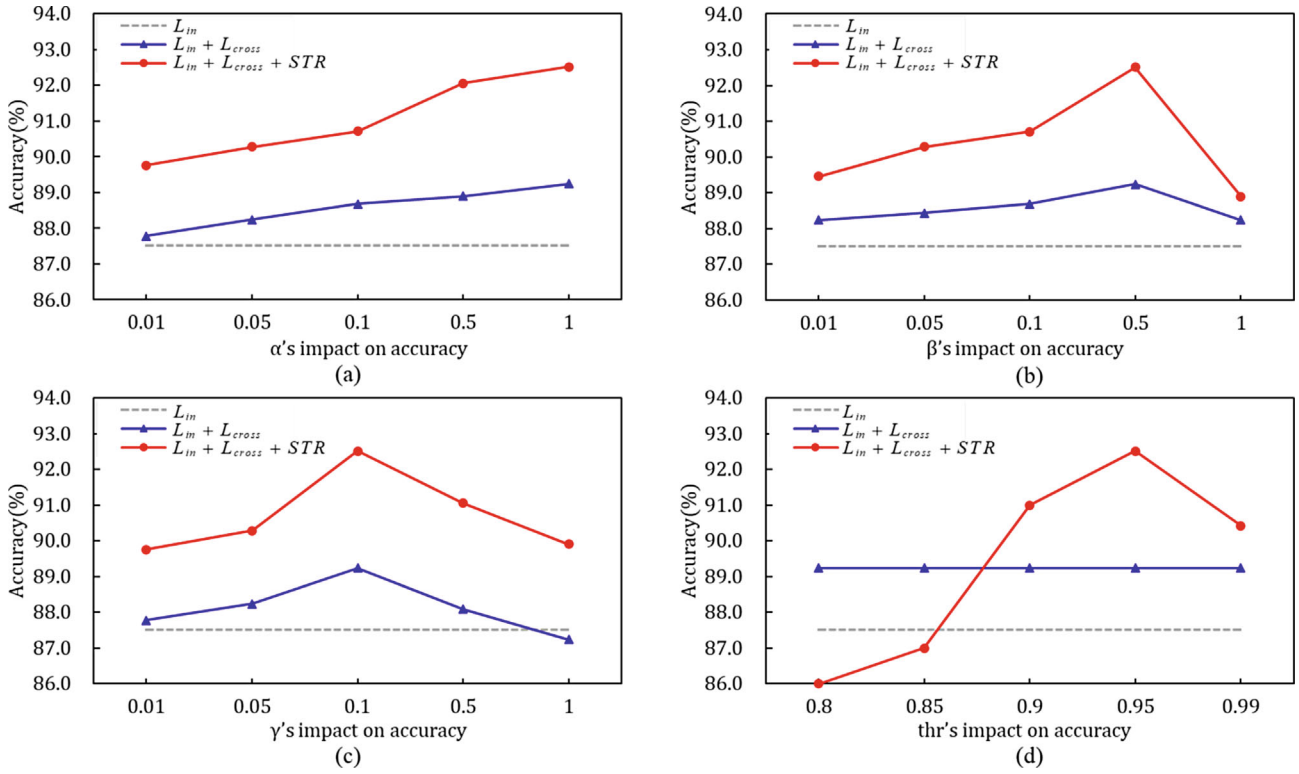


FIGURE 5 The impacts of hyperparameters on accuracy in POST framework.

to capture challenging examples from the source domain that share similarities with the target domain, thus increasing its confidence in predicting complex target domain data.

The introduction of the STR module further improves the validity of our model for the CD-FER task. It works by making the target domain data incrementally more similar to the source domain data. This is achieved by emphasizing the common similarities across both domain datasets, thereby reducing the identification requirements for the model. As shown in Table 3, the mean accuracy peaks at 70.79% upon the inclusion of the STR module (+$\mathcal{L}_{in}$+$\mathcal{L}_{cross}$+STR).

4.4.2 | Hyperparameters analysis

In our method, there are three hyperparameters, namely α, β and γ, used to adjust the contribution of $\mathcal{L}_{t\to s}$, $\mathcal{L}_{s\to t}$, and $\mathcal{L}_{ad}$ to the loss function in the domain alignment module. Another hyperparameter *thr* is the self training resampling threshold in the STR module. We conducted several sets of experimention the key hyperparameters in POST on the CK+ dataset to achieve optimal performance. By tuning, we obtained ideal hyperparameter settings. Figure 5 shows the effects of key hyperparameters on the model accuracy. From the results, it can be seen that when the value of α is adjusted from 0.01 to 1.0, the accuracy of the model show a ascending trend, while for hyperparameters β, γ and *thr* show a trend of first rising and then falling, when their value is 0.5,0.1 and 0.95, the accuracy is the highest. We also find that when *thr* is

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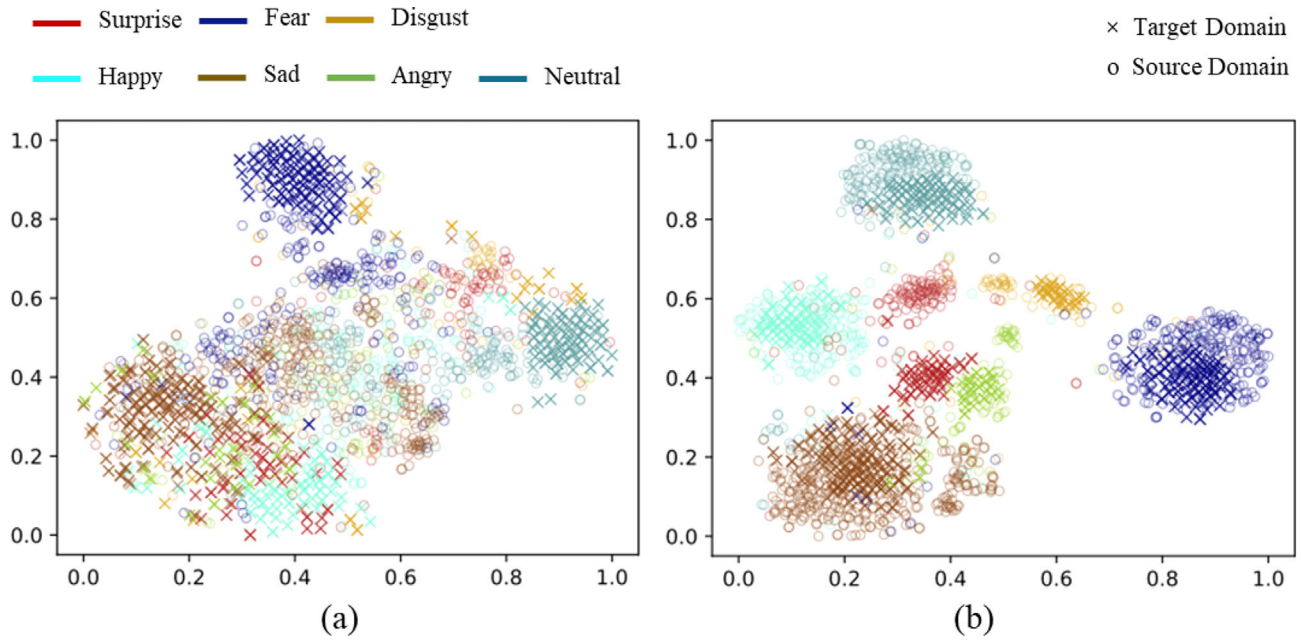


FIGURE 6 The T-SNE visualization of the feature representations learned by our proposed method before (a) and after (b) training on CK+.

below 0.9, the performance of the model drops sharply. This is because the smaller the *thr*, the easier it is for noisy images to be integrated into the resampling stage, which definitely increases the difficulty of model training. Accordingly, the hyperparameter of α, β, γ and *thr* is set to 1, 0.5, 0.1 and 0.95, respectively. We further selected the optimal hyperparameters in the POST model based on the analysis of above hyperparameters.

4.4.3 | Visualization

To verify the effect of similarity transfer, the distribution of the model extracted features was visualized using the T-SNE tool with RAF-DB as the source domain and CK+ as the target domain, and the results are presented in Figure 6. Figure 6a shows the feature distribution of the first epoch model after pre-training, while Figure 6b shows the feature distribution of the POST model after training. Seven colors represent the seven expression categories, and two symbols are used to distinguish the features from the source and target domain. Under ideal cross domain conditions, features with the same expression category should highly overlap. However, the distribution of the Figure 6a is extremely scattered, not only the overlap between the same category of source and target domain features is low, but also the classification boundary between expression categories is not clear. The Figure 6b shows that our proposed POST model can effectively improve the similarity between source and target domain features, and further improve the cross domain FER rate. The source and target domain features of the same category in Figure 6b are highly overlapping, and there is a clear classification boundary between the categories.

5 | CONCLUSIONS

In this work, we propose a prototype-oriented similarity transfer framework called POST for CD-FER to address the problem of feature distribution deviation and discriminator degradation. We first introduced bidirectional cross-attention mechanism into the ST model to focus on the similar regions in the different domain images and to get respective cross-domain features. From the perspective of domain differences, we parameterized the source and target domain prototypes to make them learnable variables, achieving domain alignment by continuously optimizing the differences between the cross-domain and domain-specific features and domain prototypes, thus effective enhancement of target

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domain discriminators. Afterwards, we constructed a self-training resampling strategy to facilitate information exchange between the source and target domains, thereby further achieving similarity transfer. We compared the performance of the proposed POST model with state-of-the-art cross-domain FER methods and show that the POST model has superior cross-domain knowledge transfer capability. However, the current CD-FER can only achieve knowledge transfer between two domain datasets in one training process. To apply it to a new target domain, repeated training is required and the efficiency is low. Therefore, how to enable the model to accurately recognize expressions on multiple target domain datasets through one training is still a research direction that needs further efforts in the future.

AUTHOR CONTRIBUTIONS

Conceptualization, Zhe Guo and Bingxin Wei; methodology, Zhe Guo; software, Bingxin Wei, Qinglin Cai and Jiayi Liu; validation, Qinglin Cai and Jiayi Liu; visualization, Bingxin Wei; funding acquisition, Zhe Guo and Yi Wang; writing–original draft preparation, Bingxin Wei; writing–review and editing, Zhe Guo; supervision, Zhe Guo and Yi Wang. All authors have read and agreed to the published version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in public repository at http://www.jeffcohn.net/Resources/. These data were derived from the following resources available in the public domain: CK+, http://www.jeffcohn.net/Resources/; FER2013, https://www.kaggle.com/datasets/msam-bare/fer2013; JAFFE, https://zenodo.org/record/3451524; SFEW2.0, https://cs.anu.edu.au/few/emotiw2015.html; RAF-DB, http://whdeng.cn/RAF/model1.html/data-set.

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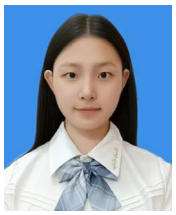


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